



Fiscal Harmony API

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Version Notes

Date	Changes
10 January 2024	Initial document version
22 March 2024	TaxMapping section updated
02 April 2024	New Customisations Section Added
05 April 2024	Addition endpoint to retrieve the list of all supported currencies on the platform
07 May 2024	Qr Code data now available in API response
15 May 2024	Additional data to associate with QR code
27 August 2024	Added HsCode to LineItem definition
04 November 2024	Added BillingStatus to Device
01 April 2025	Mandatory HsCode

Contents

Introduction	7
Resources and URIs	7
What you need to use the API	7
Authentication	7
Obtaining API Keys	8
Constructing Headers	10
Example HMACSHA256 Request Signature:	10
Optional Headers.....	10
Error Codes	11
Hours of business	11
Date Format	11
Mime Types	11
Mapping Taxes	12
Mapping Taxes.....	12
TaxMapping	12
Get All Current Tax Mappings.....	13
Input Parameters.....	13
Output Parameters	13
Get Specific Tax Mapping.....	14
Input Parameters.....	14
Output Parameters	14
Saving a new Tax Mapping.....	15
Input Parameters.....	15
Output Parameters	15
Editing an existing Tax Mapping	16
Input Parameters.....	16
Output Parameters	16
Deleting a Tax Mapping	17

Input Parameters.....	17
Output Parameters	17
Document Customisations.....	19
Customisation Object	19
Get Print Layouts	20
Input Parameters.....	20
Output Parameters	20
Get Current Customisation	21
Output Parameters	21
Saving a Customisation.....	23
Input Parameters.....	23
Output Parameters	23
Deleting a Customisation	25
Input Parameters.....	25
Output Parameters	25
Uploading a Logo.....	26
Input Parameters.....	26
Output Parameters	26
Downloading the Logo.....	27
Output Parameters	27
Mapping Currencies	28
Supported Currencies	28
Input Parameters.....	28
Output Parameters	28
Mapping Currencies.....	29
CurrencyMapping.....	29
Get All Current Currency Mappings	30
Input Parameters.....	30
Output Parameters	30

Get Specific Currency Mapping	31
Input Parameters.....	31
Output Parameters	31
Saving a new Currency Mapping	32
Input Parameters.....	32
Output Parameters	32
Editing an existing Currency Mapping.....	33
Input Parameters.....	33
Deleting a Currency Mapping.....	34
Input Parameters.....	34
Output Parameters	34
Transactions.....	35
Get Account Profile	35
Input Parameters.....	35
Output Parameters	35
Fiscal Device.....	36
Input Parameters.....	36
Output Parameters	36
CurrentConfig:.....	36
ApplicableTax	37
Submit Invoice	38
BuyerContact	38
Address	39
LineItem	39
Output Parameters	39
Submit Credit Note	40
Input Parameters.....	40
Output Parameters	40
Manually checking status	42

Input Parameters.....	43
Output Parameters	43
StatusResponse	43
Download PDF document	44
Input Parameters.....	44
Output Parameters	44
Webhooks.....	45
Server Setup for Webhooks.....	46
Save & Test Webhook URL.....	46
Delivery Failures	46
Payload.....	47
Conclusion	48

Introduction

The Fiscal Harmony API interface allows you to send invoices and credit notes through to the revenue authority for fiscal processing using your own custom Web application or user interface(s).

Resources and URIs

The Fiscal Harmony system may be accessed via a RESTful API (this document). A RESTful API utilises HTTP commands POST, GET, PUT, and DELETE in order to perform an operation on a resource at the server. This resource is addressed by a URI; and what is returned by the server is a representation of that resource depending on its current state.

What you need to use the API

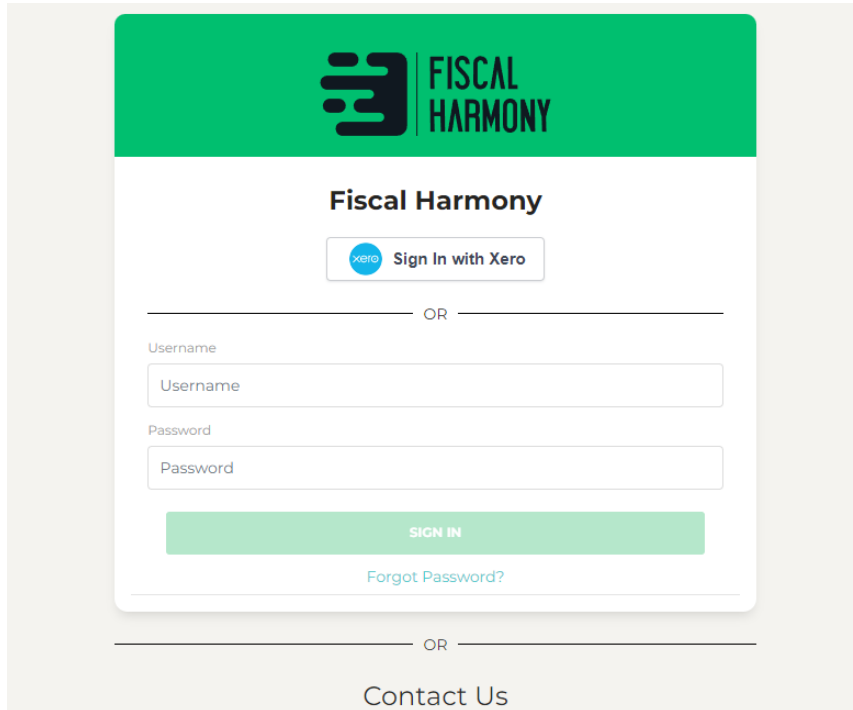
1. An active end user account on the Fiscal Harmony Platform (to create API keys)
2. The URI to access the API, referred to in this document as `{{ApiUrl}}`
3. A connected revenue authority “device” on the Fiscal Harmony account - on which transactions are processed against

Authentication

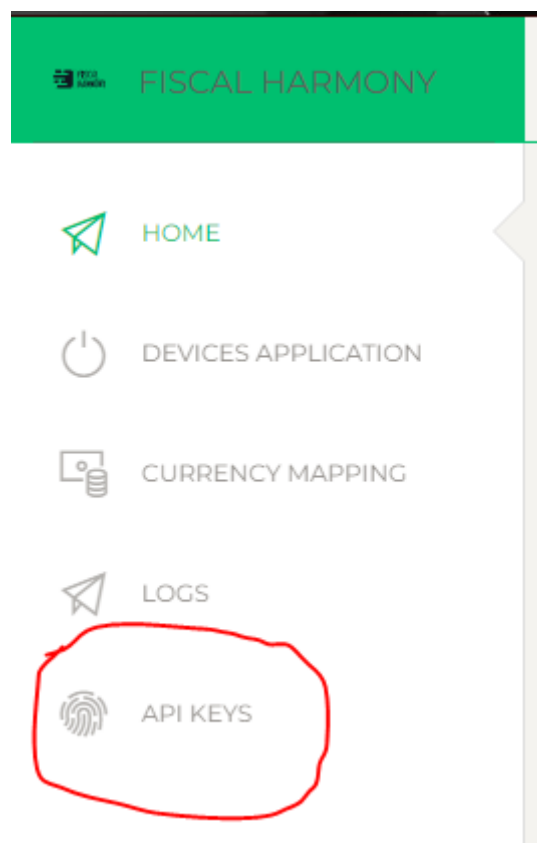
Authentication and Authorisation for the active user on the API is done through the use of HTTP Headers containing the API Key to identify the user and a request signature, signed using the API Secret. To send requests, one must perform a login request first to acquire a valid cookie to use with subsequent requests.

Obtaining API Keys

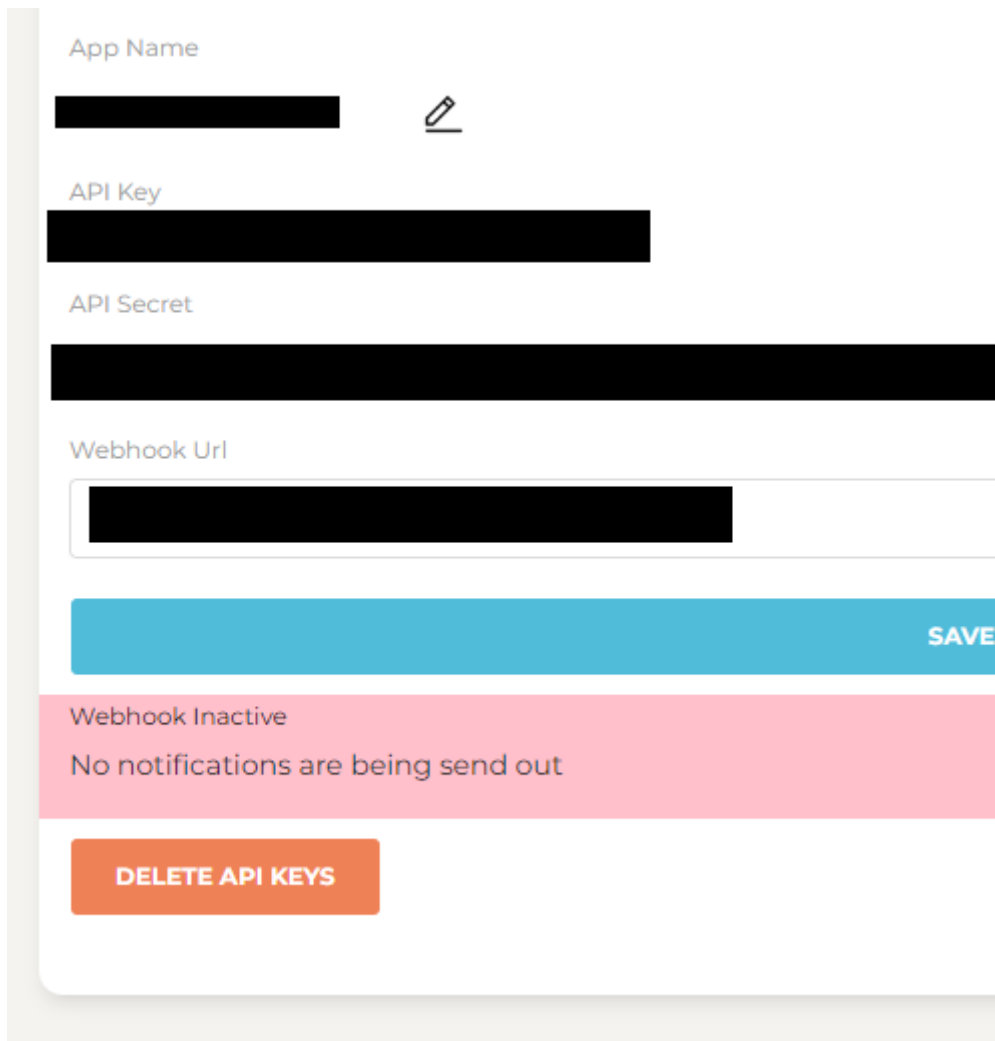
With an active account, login in and go to the API Keys menu



The image shows the Fiscal Harmony login page. At the top, there is a green header with the Fiscal Harmony logo. Below the header, the text "Fiscal Harmony" is displayed. A "Sign In with Xero" button is present, followed by a horizontal line with "OR" in the center. Below this, there are input fields for "Username" and "Password". A green "SIGN IN" button is located below the password field. A link for "Forgot Password?" is positioned below the sign-in button. At the bottom, another horizontal line with "OR" in the center is shown, followed by a "Contact Us" link.



From there Create an API Key if one is not already created:



The screenshot shows a form for creating an API key. It includes fields for 'App Name', 'API Key', 'API Secret', and 'Webhook Url'. The 'App Name' field has a text input and an edit icon. The 'API Key' and 'API Secret' fields are masked with black bars. The 'Webhook Url' field is a text input with a black bar. Below the fields is a blue 'SAVE' button. A pink notification box indicates 'Webhook Inactive' and 'No notifications are being send out'. At the bottom is an orange 'DELETE API KEYS' button.

App Name

API Key

API Secret

Webhook Url

SAVE

Webhook Inactive
No notifications are being send out

DELETE API KEYS

Once you have the API Key and Secret, you can now use them to access the API. The API Secret will be shown only once and then never again, if you lose the API Secret, you will need to login again and rotate the API Secret.

Make sure never to share the API Key and Secret and never place them where they can be accessed publicly as anyone with them can access your account the same way that you can.

Constructing Headers

To access the API, a few HTTP headers need to be present in order to be properly authenticated. Those headers are:

Header	Detail
X-Api-Key	This is the API Key obtained in the previous step.
X-Api-Signature	This is the Base64 encoded HMACSHA256 signature, signed using the API Secret, of the JSON request body for all requests with a body. Not required for GET requests
X-Application	The name of the application (provided by Fiscal Harmony).
X-App-Station	The unique identifier for the workstation/till/operator/station/user using the third party application. The third party using this API must be able to uniquely identify each operator for audit purposes. This value should not change once set.

Example HMACSHA256 Request Signature:

Let:

AS be API Secret,

Payload be the JSON data (as string) you are sending to the API,

The signature is calculated as follows:

```
using (HMACSHA256 hasher = new HMACSHA256(Encoding.UTF8.GetBytes(AS)))
{
    var hashBytes = hasher.ComputeHash(Encoding.UTF8.GetBytes(Payload));
    var signature = Convert.ToBase64String(hashBytes);
    Headers.Add("X-Api-Signature", signature);
}
```

Mandatory Headers

Though the **X-App-Station** header is mandatory. Provide it to ensure that your API connections keep working without needing to re-develop your integration. The value in this header must be unique for each user or physical device operating your application.

Error Codes

When making requests to the API, the following are the typical response codes one may get:

Code	Typical Reason
200	Success.
401	Check that API Key is correct and the document body has been correctly signed.
404	For GET requests, this means the resource that you are trying to access is not available to your account.
400	For POST requests, this means the data you sent is not in the correct format, or some critical data is missing or incorrect. The body of the message will tell you what you need to correct in order to make a successful submission
50X	The revenue authority is currently inaccessible. You may try again in a couple of minutes

Hours of business

Though some businesses operate 24 hours a day, for any other business it is strongly recommended to post transactions during normal working hours. Service outage may occur during routine evening maintenance which typically occurs outside of normal working hours.

Date Format

All dates include time and are in ISO 8601 format(https://en.wikipedia.org/wiki/ISO_8601).

Valid dates and times look like the following examples:

2025-01-08T01:04:58+02:00

2023-11-19T21:04:38+02:00

2024-12-31T15:36:01+02:00

All times are local times CAT (Central African Time) or GMT+2

Mime Types

Unless stated otherwise, all requests use the “application/json” mime type.

Mapping Taxes

In order for transactions to be processed successfully, taxes used in them must be mapped before any transaction can be processed.

Mapping Taxes

Mapping taxes ensures that a tax definition in your system correctly matches a tax definition within Fiscal Harmony. When a transaction is then posted with your system tax, it is then seamlessly translated to the corresponding Fiscal Harmony tax.

TaxMapping

The TaxMapping JSON object is in the format:

Name	Type	Mandatory	Description
Id	int	Depends	Not required on POSTing, Required on PUT, always returned on GET
UserId	int	Depends	The user profile's Id
TaxCode	string(50)	Yes	Tax code used in your application e.g. TAX001
TaxName	string(100)	No	The friendly name of the tax code to display to the user.
DestinationTaxId	int	Yes	The corresponding revenue authority tax Id to which the tax code is being mapped. This Id is the taxID of the ApplicableTax under CurrentConfig, which itself is under the GET Fiscal device endpoint. See page 37

Get All Current Tax Mappings

This endpoint returns the Fiscal Harmony account details of the authenticated user

Endpoint	/taxmapping
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
	List<TaxMapping>	Yes	List of currently available tax mappings between your application and Fiscal Harmony

Get Specific Tax Mapping

Endpoint	/taxmapping/{Id}
Type	GET

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the tax mapping to GET

Output Parameters

If found a single TaxMpapping will be returned.

Name	Type	Mandatory	Description
Id	int	Yes	The TaxMapping Id
UserId	int	Yes	The user profile's Id
TaxCode	string(50)	Yes	Tax code used in your application e.g. TAX001
TaxName	string(100)	No	The friendly name of the tax code to display to the user.
DestinationTaxId	int	Yes	The corresponding revenue authority tax Id to which the tax code is being mapped. This Id is the taxID of the ApplicableTax under CurrentConfig, which itself is under the GET Fiscal device endpoint. See page 37

Saving a new Tax Mapping

Endpoint	/taxmapping
Type	POST

Input Parameters

Name	Type	Mandatory	Description
UserId	int	No	The user profile's Id
TaxCode	string(50)	Yes	Tax code used in your application e.g. TAX001
TaxName	string(100)	No	The friendly name of the tax code to display to the user.
DestinationTaxId	int	Yes	The corresponding revenue authority tax Id to which the tax code is being mapped. This Id is the taxID of the ApplicableTax under CurrentConfig, which itself is under the GET Fiscal device endpoint. See page 37

Output Parameters

The same TaxMapping will be returned, but with a Id assigned to it

Name	Type	Mandatory	Description
Id	int	Yes	The TaxMapping Id
UserId	int	Yes	The user profile's Id
TaxCode	string(50)	Yes	Tax code used in your application e.g. TAX001
TaxName	string(100)	No	The friendly name of the tax code to display to the user.
DestinationTaxId	int	Yes	The corresponding revenue authority tax Id to which the tax code is being mapped. This Id is the taxID of the ApplicableTax under CurrentConfig, which itself is under the GET Fiscal device endpoint. See page 37

Editing an existing Tax Mapping

Endpoint	/taxmapping/{Id}
Type	PUT

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the tax mapping to be edited

Body

Name	Type	Mandatory	Description
Id	int	Yes	The Id of the TaxMapping to be edited
UserId	int	Yes	The user profile's Id
TaxCode	string(50)	Yes	Tax code used in your application e.g. TAX001
TaxName	string(100)	No	The friendly name of the tax code to display to the user.
DestinationTaxId	int	Yes	The corresponding revenue authority tax Id to which the tax code is being mapped. This Id is the taxID of the ApplicableTax under CurrentConfig, which itself is under the GET Fiscal device endpoint. See page 37

Output Parameters

Name	Type	Mandatory	Description
	boolean	Yes	Boolean value indicating success or failure of editing request

Deleting a Tax Mapping

Endpoint	/taxmapping/{Id}
Type	DELETE

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the tax mapping to be deleted

Output Parameters

None

Document Customisations

Document customisation endpoints allows the developer to specify what they want to be shown on the PDF invoice. This section allows customisation of:

- Seller contact details
- Logo that appears on invoice
- Paper size on which the invoice is to be printed
- Footer text for invoices in different currencies

Customisation Object

The Customisation JSON object is in the format:

Name	Type	Mandatory	Description
Id	int	Depends	Not required on POSTing, always returned on GET
UserId	int	No	The user profile's Id
DeviceId	int	No	The device associated with this API key. When null, this customisation is applied to all devices within the user's account When provided, this customisation applies only to the specified device. From the API, you can only set up customisations for your specific device.
InvoiceEmail	String(100)	No	The email address that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
InvoicePhone	string(20)	No	The phone number that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
PreferredPrintLayout	enum	No	Valid values are: A4 _80mmX210mm More layout options can be fetched from the Get Print Layouts endpoint on described in the following Get Print Layouts sub-section This determines what paper size is used when fiscal invoices are created.
UsdText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZwlText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency

ZarText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
BwpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
GbpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
EurText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency

Get Print Layouts

This endpoint returns the list of supported document layouts for fiscal invoice generation

Endpoint	/customisation/layout
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
	List<string>	Yes	List of currently supported document layouts. Use this in conjunction with the Customisation section to set a preferred layout for your invoices and credit notes

Get Current Customisation

Endpoint	/customisation
Type	GET

Output Parameters

If found a single Customisation object will be returned.

Name	Type	Mandatory	Description
Id	int	Yes	
UserId	int	Yes	The user profile's Id
DeviceId	int	No	The device associated with this API key. When null, this customisation is applied to all devices within the user's account When provided, this customisation applies only to the specified device.
InvoiceEmail	String(100)	No	The email address that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
InvoicePhone	string(20)	No	The phone number that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
PreferredPrintLayout	enum	No	Valid values are: A4 _80mmX210mm More layout options can be fetched from the Get Print Layouts endpoint on described in the following Get Print Layouts sub-section This determines what paper size is used when fiscal invoices are created.
UsdText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZwlText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZarText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
BwpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
GbpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
EurText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency

Saving a Customisation

This endpoint adds a customisation if it does not exist, or updates it if it already exists

Endpoint	/customisation
Type	POST

Input Parameters

Name	Type	Mandatory	Description
InvoiceEmail	String(100)	No	The email address that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
InvoicePhone	string(20)	No	The phone number that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
PreferredPrintLayout	enum	No	Valid values are: A4 _80mmX210mm More layout options can be fetched from the Get Print Layouts endpoint on described in the following Get Print Layouts sub-section This determines what paper size is used when fiscal invoices are created.
UsdText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZwlText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZarText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
BwpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
GbpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
EurText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency

Output Parameters

The same Customisation object will be returned, but with an Id assigned to it

Name	Type	Mandatory	Description
Id	int	Yes	

UserId	int	Yes	The user profile's Id
InvoiceEmail	String(100)	No	The email address that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
DeviceId	int	No	The device on which this customisation applies, which should be the device associated with this API key
InvoicePhone	string(20)	No	The phone number that shall appear under the seller name. Use this if this is different from the one initially submitted to the revenue authority
PreferredPrintLayout	enum	No	Valid values are: A4 _80mmX210mm More layout options can be fetched from the Get Print Layouts endpoint on described in the following Get Print Layouts sub-section This determines what paper size is used when fiscal invoices are created.
UsdText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZwlText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
ZarText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
BwpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
GbpText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency
EurText	string(500)	No	Footer text to add to the invoice when it uses the corresponding currency

Deleting a Customisation

Endpoint		/customisation
Type	DELETE	

Input Parameters

None

Output Parameters

None - You will receive a 200 response code if delete is successful

Uploading a Logo

The logo will appear on the fiscal invoices and credit notes using the A4 page print layout. For best visual results, it is recommended that you use an image greater than 300 pixels on each side and less than 1MB in file size. The image will be scaled as soon as you upload it to match the dimensions on the invoice/credit note. Acceptable file types are:

- png
- jpg
- jpeg

Any other file type will be rejected

Endpoint	/customisation/logo
Type	POST

Input Parameters

Form-data

Name	Type	Mandatory	Description
	File	Yes	The image to be uploaded

Output Parameters

None - You get a Success 200 status code on successful upload

Downloading the Logo

The logo may be downloaded for viewing with in your 3rd party application by using this endpoint

Endpoint	/customisation/logo
Type	GET

Output Parameters

Downloads an image file

Mapping Currencies

In order for transactions to be processed successfully, the currencies used in them must be mapped to the supported currencies before any transaction can be processed.

Supported Currencies

This endpoint returns the list of supported currencies on the Fiscal Harmony platform

Endpoint	
/currencymapping/supported-currencies	
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
	List<string>	Yes	List of currently supported currencies. Use this in conjunction with the “Mapping Currencies” section below to set up mappings with your third party application.

Mapping Currencies

Mapping currencies ensures that a currency code in your system correctly matches a currency code definition within Fiscal Harmony. When a transaction is then posted with your system currency code, it is then seamlessly translated to the corresponding Fiscal Harmony currency, for example ZWD may be mapped to ZWL or XYZ may be mapped to USD.

CurrencyMapping

The CurrencyMapping JSON object is in the format:

Name	Type	Mandatory	Description
Id	int	Depends	Not required on POSTing, Required on PUT, always returned on GET
UserId	int	Depends	The user profile's Id
SourceCurrency	string(10)	Yes	Tax code used in your application e.g. TAX001
DestinationCurrency	string(3)	No	The friendly name of the tax code to display to the user.

Get All Current Currency Mappings

This endpoint returns the Fiscal Harmony account details of the authenticated user

Endpoint	/currencymapping
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
	List<CurrencyMapping>	Yes	List of currently available currency mappings between your application and Fiscal Harmony

Get Specific Currency Mapping

Endpoint	/ urrencymapping/{Id}
Type	GET

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the currency mapping to GET

Output Parameters

If found a single CurrencyMapping will be returned.

Name	Type	Mandatory	Description
Id	int	Yes	The Currency Mapping Id
UserId	int	Depends	The user profile's Id
SourceCurrency	string(10)	Yes	Currency code used in your application e.g. ZWD
DestinationCurrency	string(3)	No	The Currency ISO code that the source currency should correspond to..

Saving a new Currency Mapping

Endpoint	/currencymapping
Type	POST

Input Parameters

Name	Type	Mandatory	Description
UserId	int	No	The user profile's Id
SourceCurrency	string(10)	Yes	Currency code used in your application e.g. ZWD
DestinationCurrency	string(3)	es	The Currency ISO code that the source currency should correspond to.

Output Parameters

The same CurrencyMapping will be returned, but with a Id assigned to it

Name	Type	Mandatory	Description
Id	int	Yes	The currency mapping Id
UserId	int	Yes	The user profile's Id
SourceCurrency	string(10)	Yes	Currency code used in your application e.g. ZWD
DestinationCurrency	string(3)	No	The Currency ISO code that the source currency should correspond to.

Editing an existing Currency Mapping

Endpoint	/currencymapping/{Id}
Type	PUT

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the currency mapping to be edited

Body

Name	Type	Mandatory	Description
Id	int	Yes	The Id of the currency mapping to be edited
UserId	int	Yes	The user profile's Id
SourceCurrency	string(10)	Yes	Currency code used in your application e.g. ZWD
DestinationCurrency	string(3)	No	The Currency ISO code that the source currency should correspond to..

Output Parameters

Name	Type	Mandatory	Description
	boolean	Yes	Boolean value indicating success or failure of editing request

Deleting a Currency Mapping

Endpoint	/currencymapping/{Id}
Type	DELETE

Input Parameters

Path variable

Name	Type	Mandatory	Description
Id	Int	Yes	The Id of the currency mapping to be deleted

Output Parameters

None

Transactions

The root of all the methods below is {{ApiUrl}}

Get Account Profile

This endpoint returns the Fiscal Harmony account details of the authenticated user

Endpoint	/profile
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
Id	int	Yes	The user profile's Id
FullName	string(100)	Yes	The name of the Fiscal Harmony Account. This typically the company name that the account represents
Username	string(50)	Yes	The login name used to enter onto the Fiscla Harmony platform
Email	string(100)	Yes	The emai address associated with the Fiscal Harmony account
FiscalProvider	string(36)	No	A unique identifier to the Fiscal Service Provider (useful, for getting support from resellers)

Fiscal Device

This endpoint returns the fiscal device assigned to the user account. A 404: Not Found error is returned if a device hasn't yet been attached to your account. If a device is available, the data will be available in the form of the output parameters shown below:

Endpoint	/fiscaldevice
Type	GET

Input Parameters

None

Output Parameters

Name	Type	Mandatory	Description
Id	int	Yes	
DeviceId	int	Yes	The Device Id assigned by the revenue authority
SerialNo	String(20)	Yes	The serial number assigned by the revenue authority
ModelName	String(10)	Yes	N/A
ModelVersion	String(10)	Yes	N/A
TIN	String(10)	Yes	The Tax Id of the owner of the Fiscal Harmony account
TaxpayerName	String(250)	Yes	The name of the taxpayer registered at the revenue authority
TradeName	String(250)	Yes	The trade name registered at the revenue authority
VATNumber	string(9)	Yes	The VAT number issued by the revenue authority
ActivatedOn	Date	Yes	The date the device was activated. Transaction dated before this date and time will not be processed
CurrentConfig	string(1000)	No	This is a JSON string, showing configuration details for this tax payer. The fields in this JSON string are detailed in the following section. Transactions will not be processed if this value is not available. Contact your Fiscal Harmony reseller to get it included/refreshed.
IsActive	boolean	Yes	States whether this device has been activated
IsLive	boolean	Yes	States whether this device is on the test platform or on the live platform.
IsSwitchedOn	boolean	Yes	States whether the device is switched on and ready for processing. Switched off devices cannot process transactions. Contact your reseller to get a switched off device turned on.
BillingStatus	int	Yes	0 - Normal 1 - Payment Due 2 - Cancelled Transactions will not be processed for any billing status that is not zero

CurrentConfig:

Name	Type	Mandatory	Description
------	------	-----------	-------------

taxPayerName	string(250)	Yes	The name of the taxpayer as registered at the revenue authority
taxPayerTIN	String(10)	Yes	The taxpayer's Tin as registered at the revenue authority
vatNumber	String(9)	No	If VAT number is available, it will be shown in this field
deviceSerialNo	String(20)	Yes	The device serial number as registered into the revenue authority system
deviceBranchName	String(250)	Yes	The branch to which this device was assigned
deviceOperatingMode	enum	Yes	"Online" means receipts/invoices/credit notes must be sent immediately after they have been created. "Offline" means transactions can be sent in batches. Useful in areas with not internet connectivity By default, account are set to "Online". If you require "Offline" capabilities, contact your Fiscal Harmony reseller.
taxPayerDayMaxHrs	int	Yes	Maximum fiscal day in hours
taxpayerDayEndNotificationHrs	int	Yes	How much time in hours before end of fiscal day that application must show notification to end user
applicableTaxes	List<ApplicableTax>	Yes	List of applicable tax rates which can be used by this taxpayer. These vary depending on changes I legislation and/or policy and so must be fetched at least once in each day.

ApplicableTax

Name	Type	Mandatory	Description
taxID	Int	Yes	Tax ID uniquely identifying a tax. This tax ID must be used in submitting invoices.
taxPercent	Decimal (5,2)	No	Tax percent. In case of exempt, field will not be returned.
taxName	String (50)	Yes	Tax name.

Submit Invoice

All invoices must be submitted on this endpoint.

Endpoint	
/invoice	
Type	POST

Input Parameters

Name	Type	Mandatory	Description
InvoiceId	String(36)	Yes	Unique invoice Id for your application. It will never be shown to the user. Recommended use of a GUID or similar global identifier.
InvoiceNumber	String(36)	Yes	Invoice Number to display on invoice
Reference	String(100)	No	Optional reference to show on invoice next to invoice number
IsTaxInclusive	boolean	Yes	Specifies whether the line items are tax inclusive or exclusive
BuyerContact	BuyerContact	Yes	Contact details for the buyer
Date	Date	Yes	The transaction date and time.
LineItems	List<LineItem>	Yes	The items on the invoice. At least one item must be provided
PaymentMethod	enum	Yes	Options available are: Cash Card MobileWallet Coupon Credit BankTransfer Other
SubTotal	decimal(19,2)	No	Total without tax
TotalTax	decimal(19,2)	No	Total Tax
Total	decimal(19,2)	No	Total with tax
CurrencyCode	string(3)	Yes	Receipt currency (ISO 4217 currency code). Currently accepted currencies are USD, ZWL, ZAR, BWP, EUR, GBP
IsDiscounted	boolean	No	Defaults to false. If any line has a discount but this is not set, the discount will be ignored. If any line item has a discount, specify this value as true
IsRetry	boolean	No	Set this only if a transaction failed, and your are retrying it. This allows an invoice older than 24 hours to be processed, provided that its processing failed before

BuyerContact

Contact details for the buyer

Name	Type	Mandatory	Description
Name	string(250)	Yes	The name of the buyer or registered tax payer name at revenue authority
TradeName	String(250)	No	The buyer's registered trade name
Tin	string(10)	No	The buyer's TIN
VatNumber	string(10)	No	The buyer's VAT number

Phone	String(20)	No	
Email	String(100)	No	
Address	Address	No	

Address

Name	Type	Mandatory	Description
HouseNo	string(100)	Yes	House number
Street	string(100)	Yes	Street, stand number, village
City	string(100)	Yes	City, town, growth point, farming area, mining area
Province	string(100)	Yes	Province name

LineItem

Name	Type	Mandatory	Description
Description	string(200)	Yes	Product or service name
TaxCode	string(50)	Yes	The tax code for this line item. Ensure the tax code you use is mapped first before sending an invoice with it, otherwise the invoice processing will fail
ProductCode	string(8)	Yes	Provides the HsCode specification of the line item. This field is for use on HsCodes as provided by the revenue authority
UnitAmount	decimal(19,3)	Yes	The unit price of the product or service
Quantity	decimal(19,3)	Yes	The quantity of the product or service units
LineAmount	decimal(19,2)	Yes	The total amount for the line item Validations - If DiscountAmount is not provided, this should be equal to UnitAmount*Quantity - If DiscountAmount >0, this field should be equal to ((UnitAmount*Quantity) - DiscountAmount)
DiscountAmount	decimal(19,2)	No	If a discount is being given, this is the discount amount. It is only considered if IsDiscounted is true on the invoice object

Output Parameters

Name	Type	Mandatory	Description
	string	Yes	Simple JSON string. This is a request Id that you must use to correlate with the status produced by a webhook event or for fetching results using the "status" endpoint described in the next sections of this document

Submit Credit Note

Credit notes must be provided on this endpoint

Endpoint	/creditnote
Type	POST

Input Parameters

Name	Type	Mandatory	Description
CreditNoteld	String(36)	Yes	Unique credit note Id for your application. It will never be shown to the user. Recommended use of a GUID or similar global identifier.
OriginalInvoiceld	String(36)	Yes	The Invoiceld of the original invoice on which this credit note is being applied
CreditNoteNumber	String(36)	Yes	Credit note number to show on the fiscal credit note
Reference	string(250)	Yes	The reason why the credit note is being created.
IsTaxInclusive	boolean	Yes	Specifies whether the line items are tax inclusive or exclusive
BuyerContact	BuyerContact	Yes	Contact details for the buyer
Date	Date	Yes	Transaction date and time
LineItems	List<LineItem>	Yes	The items on the credit note. At least one item must be provided
PaymentMethod	string	Yes	Options available are: Cash Card MobileWallet Coupon Credit BankTransfer Other
SubTotal	decimal(19,2)	No	Total without tax
TotalTax	decimal(19,2)	No	Total tax
Total	decimal(19,2)	No	Total with tax
CurrencyCode	string	Yes	Receipt currency (ISO 4217 currency code). Currently accepted currencies are USD, ZWL, ZAR, BWP, EUR, GBP
IsRetry	boolean	No	Set this only if a transaction failed, and you are retrying it. This allows an credit notes older than 24 hours to be processed, provided that its processing failed before

Output Parameters

Name	Type	Mandatory	Description
	string	Yes	Simple JSON string. This is a request Id that you must use to correlate with the status produced by a webhook event or for fetching results using the "status" endpoint describes in the next sections of this document

Manually checking status

This point is used to manually check of get the result of transactions with the provided transaction Ids. You will only have access to transactions processed in the last seven (7) days using this endpoint. Only use this method if you have lost some webhook events and you want to recover the details of the missed events.

The recommended way to get the status of a transaction is to use webhooks.

Webhooks and how to implement them is detailed in the next chapter.

Endpoint		/status
Type		POST

Input Parameters

Name	Type	Mandatory	Description
	List<string>	Yes	List of request Ids to check status for. Only 100 request Ids are processed at a time. Make sure you do not send more than 100 request Ids in order to get consistent results.

Output Parameters

Name	Type	Mandatory	Description
	List<StatusResponse>	Yes	

StatusResponse

Name	Type	Mandatory	Description
RequestId	string	Yes	The request identifier. It was initially sent back at the time of submission
Success	boolean	Yes	States where the fiscal process has been successful
Error	string	No	Only sent back if Success=false. This provides a client-friendly message as to why the processing failed and/or what they can do about it to correct it.
IsActionable	boolean	No	Only sent back if Success=false. State whether the submission should be tried again after correcting the error. When IsActionable=false, the submission cannot be corrected, subsequent submissions of the same invoice/credit note will lead to the same error.
FiscalInvoicePdf	string	No	Only sent back if Success=true. This is the filename reference to use to download the PDF file using the download endpoint.
QrData	QrDataObject	No	If success is true, this will contain the data used to create the QR Code for the invoice/credit note

QrDataObject

Name	Type	Mandatory	Description
QrCodeUrl	string(100)	Yes	The Url in the QR Code. Use this to create the QrCode to display on your receipt
VerificationCode	string(20)	Yes	The verification code for your receipt, this must be placed need the QrCode with the instruction text to for the buyer to be able to verify the receipt
FiscalDay	int	Yes	The current fiscal day number
DeviceId	int	Yes	The virtual Device ID on the Revenue Authority system
InvoiceNumber	int	Yes	The document/invoice number on the Revenue authority system

Download PDF document

This endpoint is used to retrieve or download a generated PDF document (invoice or credit note), provided you have the filename - which is returned in the field `FiscalInvoicePdf` in a webhook event or via a manual transaction status check.

Endpoint	/download/{filename}
Type	GET
Returned Mime type	application/pdf

Input Parameters

None

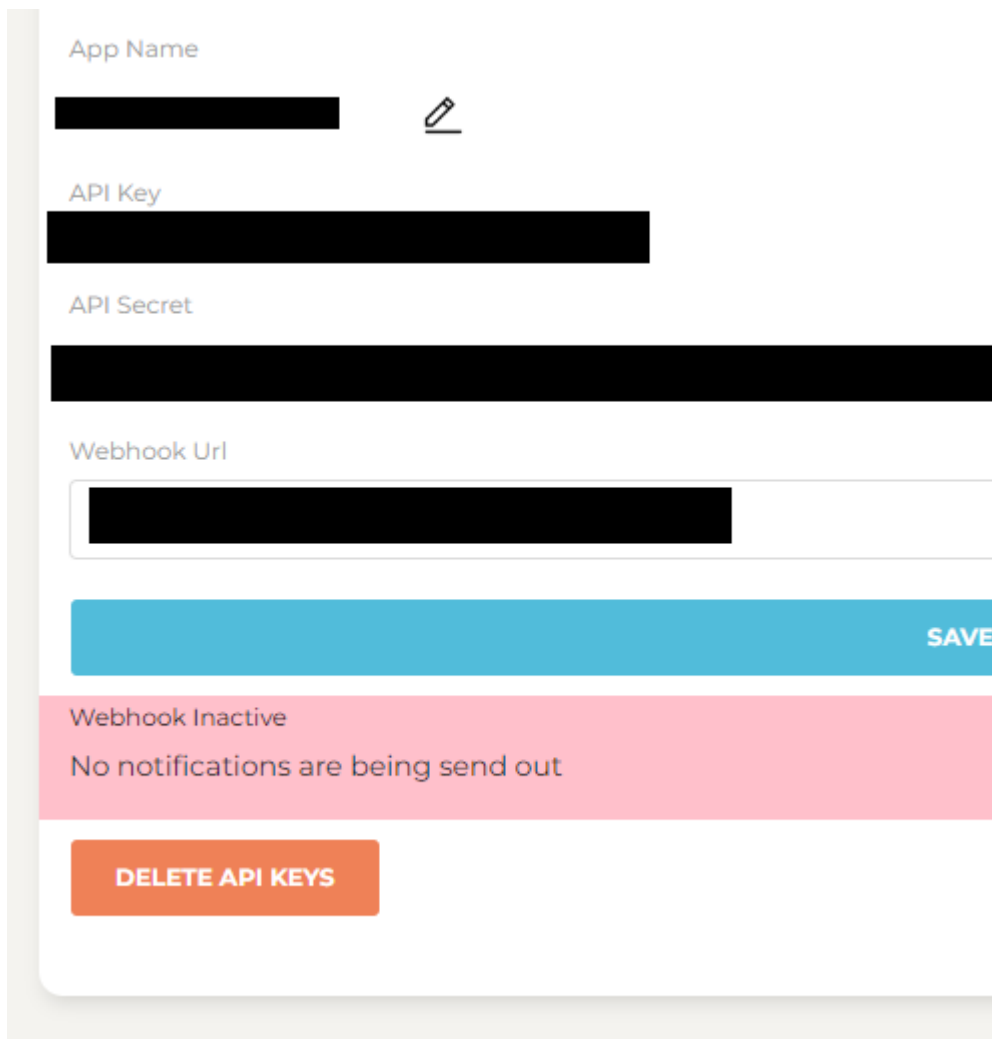
Output Parameters

Downloads a PDF document file

Webhooks

Webhooks allow you to subscribe to invoice/credit note processing events that happen in Fiscal Harmony. When one of these events are triggered we'll send a HTTP POST payload to the webhook's configured URL.

Webhooks can be created on all API Keys. The image below shows where to enter the webhook url of your choice - alongside the API KEY and Secret:



The screenshot shows a web form for configuring a webhook. It includes input fields for 'App Name', 'API Key', 'API Secret', and 'Webhook Url'. The 'App Name' field has a small edit icon to its right. Below the input fields is a blue 'SAVE' button. Underneath the 'SAVE' button is a pink notification box with the text 'Webhook Inactive' and 'No notifications are being send out'. At the bottom of the form is an orange button labeled 'DELETE API KEYS'.

Once you have entered a valid HTTPS url, click “Save & Test”. Test data is sent to your webhook URL and the webhooks are activated on receiving responses which have been correctly processed. The status will change to “Webhook Active” and you will start receiving events on your URL.

Server Setup for Webhooks

Once you've created the webhook you'll need to set up your server to receive the HTTP POST requests at the endpoint you specified. There are just a few basic things that we expect from your endpoint

1. It uses HTTPS on the standard 443 port
2. It responds within 5 seconds with a 2xx status code
3. There are no cookies in the response headers
4. If the signature is invalid a 401 Unauthorised status code is returned

If we don't receive an acceptable response when sending events we will continue retrying the request, with decreasing frequency for up to 24 hours. After 24 hours the webhook will be disabled and needs to be re-enabled in the online user account.

If you want to test webhooks in your local dev environment, Github have a [handy guide](#) using Sinatra and ngrok.

Save & Test Webhook URL

To ensure the requests you receive are coming from Fiscal Harmony you need to verify the signature provided in the **X-Api-Signature** header. When creating or re-enabling a webhook subscription (or updating the subscription url) the user will be prompted to 'Save & Test Webhook URL' validation. This validation process will be a series of HTTPS POST requests to the url provided in the Webhook Url field.

If the payload is hashed using HMACSHA256 with your API Secret and base64 encoded, it should match the signature in the header. This is a **correctly signed payload**. If the signature does not match the hashed payload it is an **incorrectly signed payload**.

To gain "Save & Test Webhook URL" validation, the receiving url must respond with a 2xx status for all **correctly signed payloads** and status: 401 Unauthorized for all **incorrectly signed payloads**.

It must also respond within the guidelines of all webhook requests listed above.

If the validation is passed, the status displayed in the console will change from **Webhook Inactive** to **Webhook Active**. If the validation fails, the status will return to **Webhook Inactive**.

Delivery Failures

When you fail to respond correctly to a webhook we will immediately retry. Any subsequent retry attempts will exponentially back-up at 2 minutes, 20 minutes, 2 hours until a successful response is given.

If after 24 hours we still have not received a successful response we will disable your webhook subscription.

Once the issue has been resolved you can re-enable your webhook subscription and complete the “Save & Test Webhook URL” process to continue receiving your webhooks. Any events that happen while your subscription is in the Retry or Disabled states will be saved for up to 24 hours and replayed in order once your subscription is healthy again.

Payload

The webhook event will contain a list of at least one or more processing results.

Each processing result will have the following fields:

Name	Type	Mandatory	Description
	List<StatusResponse>	Yes	

StatusResponse:

Name	Type	Mandatory	Description
RequestId	string	Yes	The request identifier. It was initially sent back at the time of submission
Success	boolean	Yes	States where the fiscal process has been successful
Error	string	No	Only sent back if Success=false. This provides a client-friendly message as to why the processing failed and/or what they can do about it to correct it.
IsActionable	boolean	No	Only sent back if Success=false. State whether the submission should be tried again after correcting the error. When IsActionable=false, the submission cannot be corrected, subsequent submissions of the same invoice/credit note will lead to the same error.
FiscalInvoicePdf	string	No	Only sent back if Success=true. This is the filename reference to use to download the PDF file using the download endpoint.
QrData	QrDataObject	No	If success is true, this will contain the data used to create the QR Code for the invoice/credit note

QrDataObject

Name	Type	Mandatory	Description
QrCodeUrl	string(100)	Yes	The Url in the QR Code. Use this to create the QrCode to display on your receipt
VerificationCode	string(20)	Yes	The verification code for your receipt, this must be placed need the QrCode with the instruction text to for the buyer to be able to verify the receipt
FiscalDay	int	Yes	The current fiscal day number
DeviceId	int	Yes	The virtual Device ID on the Revenue Authority system
InvoiceNumber	int	Yes	The document/invoice number on the Revenue authority system

Conclusion

This API is under heavy active development, changes will be kept to a minimum, but any breaking changes will be communicated in advance in order for stakeholders to make the necessary changes. Transition routes will be provided to cater for complex changes.

Overall the expectation is low that the structure of the API will change, though additional security requirements may be implemented in the future.

Any questions on this API document, address them to support@fiscalharmony.co.zw and a support liaison will be assigned to you to offer you assistance.

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